

CFB PETAWAWA, ON, Canada

WMO: 716250

Lat: 45.948N

Lon: 77.311W

Elev: 130

StdP: 99.77

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-25.1	-33.4	0.2	-27.8	-30.3	0.2	-24.4	9.4	-10.6	8.4	-10.4	1.2	280	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	30.8	21.5	28.9	20.5	27.1	19.8	23.1	28.5	22.0	26.8	21.0	25.1	3.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.4	25.4	20.5	15.4	24.3	19.5	14.5	23.3	69.3	28.5	65.2	26.7	61.6	25.1	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.3	-33.0	33.8	3.0	1.7	-35.2	35.0	-36.9	36.0	-38.6	37.0	-40.8	38.2		
(5)			-33.2	25.0	2.9	1.0	-35.3	25.7	-37.0	26.3	-38.7	26.9	-40.8	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-11.4	-9.4	-3.4	4.8	12.2	17.3	19.8	18.8	14.6	7.8	0.6	-6.7
	DBStd	12.07	7.67	6.72	6.38	4.69	4.49	3.82	3.09	3.18	4.29	4.64	5.22	6.87
	HDD10.0	2729	664	543	417	168	27	1	0	0	8	101	283	518
	HDD18.3	4855	922	777	674	407	196	63	18	33	128	328	532	777
	CDD10.0	1085	0	0	1	11	96	221	305	273	146	32	1	0
	CDD18.3	169	0	0	0	0	7	34	65	48	15	1	0	0
	CDH23.3	1662	0	0	1	12	113	344	636	421	127	9	0	0
	CDH26.7	473	0	0	0	3	29	104	194	113	28	1	0	0
Wind	WSAvg	2.6	2.7	2.8	3.0	3.1	2.7	2.4	2.2	2.1	2.3	2.8	2.9	2.8
Precipitation	PrecAvg	851	53	46	65	65	71	89	74	83	82	74	71	75
	PrecMax	1154	102	112	118	101	135	224	141	165	165	141	114	154
	PrecMin	682	18	5	27	16	19	42	36	25	43	32	32	30
	PrecStd	111	19	26	22	26	35	43	28	43	30	30	25	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	7.1	14.6	25.0	29.8	32.2	33.5	32.7	29.9	24.1	15.6	9.3
		MCWB	6.3	4.0	8.2	15.6	18.8	21.7	23.4	22.5	22.5	16.9	12.0	7.4
	2%	DB	3.7	3.8	10.1	19.7	26.5	29.7	30.9	29.6	26.6	20.4	12.8	6.0
		MCWB	2.7	2.0	6.0	11.3	16.9	20.9	21.8	21.0	20.4	15.5	10.3	4.8
	5%	DB	1.7	2.3	7.3	16.0	23.6	27.3	28.8	27.5	24.4	17.7	10.1	3.4
		MCWB	0.7	0.7	3.8	9.4	15.5	19.4	20.5	20.3	19.0	14.0	8.0	2.3
	10%	DB	-0.1	0.7	5.1	12.9	21.0	24.9	26.9	25.6	22.2	15.3	7.8	1.9
		MCWB	-0.9	-0.4	2.5	7.1	14.3	18.3	20.0	19.4	17.9	12.3	6.0	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	4.5	10.9	16.5	21.0	23.7	24.9	23.8	23.3	18.8	13.1	7.7
		MCDB	6.7	5.8	13.7	22.4	27.2	29.7	31.0	29.5	28.0	22.2	14.7	8.8
	2%	WB	2.8	2.2	6.5	13.0	18.8	22.2	23.3	22.5	21.4	16.3	10.5	5.0
		MCDB	3.5	3.3	9.4	17.8	24.1	27.7	28.6	27.4	25.1	19.4	12.2	6.0
	5%	WB	0.7	1.0	4.1	9.9	17.0	20.8	22.0	21.4	20.0	14.5	8.2	2.6
		MCDB	1.5	1.9	6.7	14.8	21.4	25.1	26.6	25.7	23.2	17.1	9.8	3.3
	10%	WB	-1.1	-0.4	2.5	8.0	15.1	19.4	21.0	20.4	18.4	12.6	6.1	1.0
		MCDB	-0.2	0.9	4.8	11.8	19.3	23.4	25.0	24.1	21.4	15.0	7.6	1.7
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.6	11.2	11.6	12.9	12.2	12.4	12.1	11.9	9.5	8.0	8.3
		MCDBR	8.7	10.3	13.1	18.5	18.1	15.5	15.1	14.7	14.7	13.9	10.9	7.9
		MCWBR	8.2	9.0	9.2	11.0	9.5	7.7	7.3	7.5	8.7	9.2	8.9	7.2
	5% WB	MCDBR	8.6	9.1	11.5	15.6	14.2	12.7	12.7	12.3	12.4	12.6	10.0	7.6
		MCWBR	8.4	8.3	8.6	10.4	8.0	7.2	6.9	6.8	7.7	8.9	8.6	7.1
Clear-Sky Solar Irradiance	taub	0.270	0.287	0.321	0.355	0.372	0.412	0.407	0.394	0.371	0.344	0.310	0.273	
	taud	2.349	2.290	2.338	2.375	2.385	2.293	2.322	2.356	2.414	2.475	2.491	2.393	
	Ebn at Noon	839	891	908	903	895	856	855	853	844	819	788	795	
	Edh at Noon	74	96	107	113	116	128	123	115	99	81	66	64	
All-Sky Solar Radiation	RadAvg	1.35	2.30	3.67	4.56	5.16	5.69	5.64	4.93	3.83	2.25	1.32	0.98	
	RadStd	0.19	0.28	0.33	0.39	0.46	0.42	0.47	0.28	0.36	0.23	0.15	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page